

based on Processing and other open-source software.

PHP. PHP (recursive acronym for PHP: Hypertext Preprocessor) is a widely-used open source general-purpose scripting language that is especially suited for web development and can be embedded into HTML. It

is especially suited to server-side web development, in which case PHP generally runs on a web server. Any PHP code in a requested file is executed by the PHP runtime, usually to create dynamic web page content or dynamic images used on websites or elsewhere. PHP5 is used in this project to run and edit default webpage.

Wireless IMU

app. This app (Fig. 16) measures and reports a body's specific force, angular rate and sometimes the magnetic field surrounding the body, using a combination of accelerometers and gyroscopes, and sometimes magnetometers. An inertial measurement unit (IMU) works by detecting the current rate of acceleration using one or more accelerometers, and detects changes in rotational attributes like pitch, roll and yaw using one or more gyroscopes. And some also include a magnetometer, mostly to assist calibration against orientation drift.

The accelerometers are placed such that their measuring axes are orthogonal to each other. Three gyroscopes are placed in a similar orthogonal pattern, measuring rotational position in reference to an arbitrarily chosen coordinate system. Magnetometers allow better performance for dynamic orientation calculation in attitude and heading reference systems.

Dual screen. The DualScreen app by Goestoweb allows the wire of the smart-

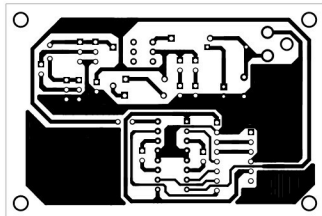


Fig. 17: Actual-size PCB layout of virtual telepresence robot

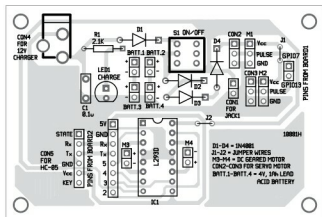


Fig. 18: Components layout of the PCB

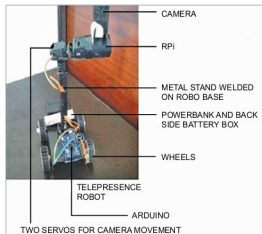


Fig. 19: Author's prototype

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

phone to be shown in two different parts. This is done to get accurate visuals through the VR lens. There are two eyepieces in VR headset, hence dual-screen mode is used.

Bluetooth Electronics. This application by Keuwlsoft has to be installed in the smartphone. It allows the user to drive the motor from the smartphone. The directions of the robot are sent via Bluetooth.

Construction and testing

An actual-size PCB layout of the virtual telepresence circuit is shown in Fig. 17 and its components layout in Fig. 18. Mount the components on the PCB, connect the boards (Board1 and Board2) and the four batteries. Use a suitable chassis for the robot. Fix all the batteries underneath the chassis. Mount the PCB and boards on top of the chassis. Connect servo motors M1 and M2. Attach the RPi camera on these two motors for horizontal and vertical movements.

Connect the two DC geared motors to the chassis of the robot for moving forward, backward, left and right.

When hardware assembly is done, install all the required apps in your smartphone as mentioned earlier. Enable dual screen mode. Place your smartphone on the VR headset such as a VR Box. Now, put the VR headset on your head. Adjust the right and left lenses of the VR glasses to make sure that the image is being displayed properly. If everything is fine, turn your head left and right, the camera fitted on the robot will also move in the same direction accordingly. You can also control the robot (without VR) using Bluetooth Electronics app. The author's prototype is shown in Fig. 19. **EFY**

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